

differ on a set of null measure). So we have that $\nabla w_1(x) = \nabla w_2(x) = 0$ for a.a. $x \in \Omega \setminus \tilde{\Omega}$. We may also write that $\frac{\nabla w_1(x)}{w_1(x)} = \frac{\nabla w_2(x)}{w_2(x)} = 0$ for a.a. $x \in \Omega \setminus \tilde{\Omega}$.

Fact II: For a.a. $x \in \tilde{\Omega}$ we have that $\frac{\nabla w_1(x)}{w_1(x)} = \frac{\nabla w_2(x)}{w_2(x)}$.

Since $x \in \tilde{\Omega}$ we have that $\nabla w_1(x), \nabla w_2(x) \neq 0$. But to have equalities in (43) we deduce that $|\nabla w_1(x)| \cdot |\nabla w_2(x)| = \nabla w_1(x) \cdot \nabla w_2(x)$. This means that there is some $\lambda(x) > 0$ such that $\nabla w_2(x) = \lambda(x) \nabla w_1(x)$. Following the cases in which equality can occur in Lemma 4.1, having $r > 1$ and $|\nabla w_1(x)|, |\nabla w_2(x)| > 0$, we are in the case (iii) from which we can write that: $\frac{|\nabla w_1(x)|}{w_1(x)} = \frac{|\nabla w_2(x)|}{w_2(x)} \implies \frac{|\nabla w_1(x)|}{w_1(x)} = \frac{\lambda(x) |\nabla w_1(x)|}{w_2(x)} \implies \lambda(x) = \frac{w_2(x)}{w_1(x)}$. So the claim is proved, since $\nabla w_2(x) = \frac{w_2(x)}{w_1(x)} \nabla w_1(x)$.

Fact III: The function $\frac{w_2}{w_1}$ is constant.

At this moment we have that $\frac{\nabla w_1(x)}{w_1(x)} = \frac{\nabla w_2(x)}{w_2(x)}$ a.e. on the entire domain Ω . Using now Lemma 5.1 proved at the end of this proof, we deduce that $\frac{w_2}{w_1} \in W_{\text{loc}}^{1,1}(\Omega)$ and: $\nabla \left(\frac{w_2}{w_1} \right) = \frac{w_2}{w_1} \left(\frac{\nabla w_2}{w_2} - \frac{\nabla w_1}{w_1} \right) = 0$ a.e. on Ω . Since Ω is a connected domain we infer that $\frac{w_2}{w_1}$ is a constant function. Let $w_2 = \lambda w_1$ a.e. on Ω for some $\lambda > 0$. Substituting this in (54) we obtain a.e. on Ω that:

$$\begin{aligned} \mathbf{a}(x, \nabla w_1) \cdot \nabla (w_1 - \lambda^r w_1) &= \mathbf{a}(x, \lambda \nabla w_1) \cdot \nabla \left(\frac{w_1}{\lambda^{r-1}} - \lambda w_1 \right) \\ \iff (1 - \lambda^r) \mathbf{a}(x, \nabla w_1) \cdot \nabla w_1 &= \frac{1 - \lambda^r}{\lambda^{r-1}} \mathbf{a}(x, \lambda \nabla w_1) \cdot \nabla w_1 \\ \iff (1 - \lambda^r) \lambda^{r-1} \Phi(x, |\nabla w_1|) |\nabla w_1| &= (1 - \lambda^r) \Psi(x, \lambda |\nabla w_1|) \lambda \nabla w_1 \cdot \nabla w_1 \\ \iff (1 - \lambda^r) \lambda^{r-1} \Phi(x, |\nabla w_1|) |\nabla w_1| &= (1 - \lambda^r) \Phi(x, \lambda |\nabla w_1|) |\nabla w_1| \\ \iff (1 - \lambda^r) |\nabla w_1| [\Phi(x, \lambda |\nabla w_1|) - \lambda^{r-1} \Phi(x, |\nabla w_1|)] &= 0. \end{aligned}$$

Therefore we may have $\lambda = 1$ and the equality holds. If $\lambda \neq 1$ then $\nabla w_1(x) = 0$ or $\Phi(x, \lambda |\nabla w_1|) = \lambda^{r-1} \Phi(x, |\nabla w_1|)$ for a.a. $x \in \Omega$. This disjunction is equivalent to the second relation since $\Phi(x, 0) = 0$.

If we know that the function $(0, \infty) \ni s \mapsto \frac{\Phi(x, s)}{s^{r-1}}$ is **strictly increasing** for a.a. $x \in \Omega$ (i.e. **(H7')** holds), then equality can occur when $\lambda = 1$ (i.e. $w_1 \equiv w_2$). If $\lambda \neq 1$ then for $x \in \tilde{\Omega}$ (i.e. $\nabla w_1(x) \neq 0$) we have that $\Phi(x, \lambda |\nabla w_1(x)|) = \lambda^{r-1} \Phi(x, |\nabla w_1(x)|)$ which rewrites as $\frac{\Phi(x, |\nabla w_1(x)|)}{|w_1(x)|^{r-1}} = \frac{\Phi(x, \lambda |\nabla w_1(x)|)}{(\lambda |w_1(x)|)^{r-1}}$. This equality

is impossible to hold from the strict monotony of the function involved there. So $|\tilde{\Omega}| = 0$ which means that $\nabla w_1 = 0$ a.e. on Ω . Therefore $\nabla w_2 = \lambda \nabla w_1 = 0$ a.e. on Ω . Finally, since Ω is connected we have that w_1 and w_2 are different constant functions. The theorem is now completely proved. $\square$

Now we shall state and prove the lemma we have used above.

Lemma 5.1. *Let $\Omega \subset \mathbb{R}^N$ be an open, bounded and connected domain. If $w_1, w_2 \in W^{1,p(x)}(\Omega)$ with:*

- (i) $w_1, w_2 > 0$ a.e. on Ω ;
- (ii) $\frac{w_1}{w_2}, \frac{w_2}{w_1} \in L^\infty(\Omega)$;
- (iii) $\frac{\nabla w_1}{w_1}, \frac{\nabla w_2}{w_2} \in L_{\text{loc}}^1(\Omega)^N$.

Then $\frac{w_2}{w_1}, \frac{w_1}{w_2} \in W_{\text{loc}}^{1,1}(\Omega)$ and moreover:

$$\nabla \left(\frac{w_2}{w_1} \right) = \frac{w_1 \nabla w_2 - w_2 \nabla w_1}{w_1^2} = \frac{w_2}{w_1} \left(\frac{\nabla w_2}{w_2} - \frac{\nabla w_1}{w_1} \right). \quad (55)$$